Longwave Infrared Hyperspectral Emissivity Retrieval With Robustness to Spectral Uncertainty

Nicholas Durkee, Joshua N. Ash, and Joseph Meola

***Abstract*— Exploitation of hyperspectral long-wave infrared (LWIR) data often involves an emissivity retrieval process where ground-surface emissivity is estimated from at-sensor radiance measurements. Emissivity retrieval consists of both atmospheric compensation and temperature separation, however, the presence of spatial–spectral distortion or small spectral calibration errors can engender large errors in emissivity retrieval. In this article, we propose an efficient emissivity retrieval algorithm that jointly performs atmospheric compensation and temperature-emissivity separation while compensating for spectral uncertainty that may exist due to smile or calibration artifacts. Without relying on particular in-scene materials, such as blackbodies or in-library elements, the proposed algorithm estimates unique spectral response function (SRF) band centers and bandwidths at each spatial-spectral index of the focal plane array (FPA) in order to compensate for optical distortions. Like the automatic retrieval of temperature and emissivity using spectral smoothness (ARTEMISS) method, the approach utilizes a spectral smoothness form of regularization. The proposed algorithm, decoding infrared atmospheres in ARTEMISS (DIANA), relies on a deep neural network decoder in order to efficiently model a sequence of SRF-weighted transmission-upwelling-downwelling (TUD) vectors applicable across the FPA and employs a genetic algorithm to perform the high-dimensional inference task. Experimental results on synthetic and measured data quantify performance in a range of environmental conditions and distortion levels, while results from measured data highlight a 26%–31% average reduction in emissivity estimation error relative to existing algorithms.**

***Index Terms*— Atmospheric compensation (AC), hyperspectral imaging, machine learning, spectral calibration.**

I. Introduction

HYPERSPECTRAL imagery is characterized by an abundance of observation bands within the electromagnetic spectrum. The availability of this rich spectral information is well-suited to a number of remote sensing applications, such as material classification, abundance estimation, change detection, and anomaly detection. However, the spectral signatures transduced by a hyperspectral sensor are not solely functions of the materials in the scene, but also depend on local environmental conditions such as the atmosphere and ground temperature.

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To facilitate sensor exploitation tasks in the long-wave infrared (LWIR) domain, a material's spectral emissivity, $\boldsymbol{\epsilon}$, may be considered. Emissivity is an intrinsic property that quantifies a material's thermal radiance relative to a blackbody emitter, thus providing a spectral signature independent of measurement conditions. Accordingly, it is often desirable to perform LWIR data analysis in the emissivity domain in order to avoid temporally and spatially dependent environmental effects. In this article, we focus on LWIR emissivity estimation in the presence of multiple nuisance quantities, including unknown atmospheric conditions, unknown ground surface temperatures, and unknown deviations of sensor band center and bandwidth from assumed truth.

For a sensor with N_b bands, emissivity estimation is an ill-posed problem, even with perfect sensor calibration and known atmospheric conditions. This is because, for each pixel, N_b emissivity values and one ground surface temperature ($N_b + 1$ unknowns) must be estimated from N_b radiance measurements [1]. Various temperature-emissivity separation (TES) algorithms with associated regularization strategies have been proposed to retrieve emissivities in this ill-posed problem. In [2], TES was posed as a Bayesian estimation problem where prior distributions on emissivity structure and temperature were used to disambiguate candidate solutions. Regularization methods requiring fewer *a priori* assumptions include fixing the emissivity of a single band [3], exploitation of spectral smoothness [4], exploitation of the approximate linearity of self-emissions over narrow bands [5], utilizing an approximation of the relationship between brightness temperature and emissivity [6], and restricting emissivity vectors to lie in a linear subspace [7]. A minimal-assumption approach to TES based on maximum entropy estimation was considered in [8], while recent work [9] considers the advantage of time-series observations.

When atmospheric conditions are unknown, atmospheric compensation (AC) methods must be combined with TES strategies in order to retrieve emissivity. In-scene atmospheric compensation (ISAC) [10] produces atmospheric estimates via regression analysis of in-scene blackbodies, while the empirical line method [11] relies on multiple known targets to estimate atmospheric parameters via least-squares. Utilizing relative transmittance and path radiance metrics between strong and weak absorption channels, the autonomous atmospheric compensation (AAC) [12] algorithm estimates AC profiles. The fast line-of-sight atmospheric analysis of spectral hypercubes–infrared (FLAASH-IR) algorithm [13] and the automatic retrieval of temperature and emissivity using

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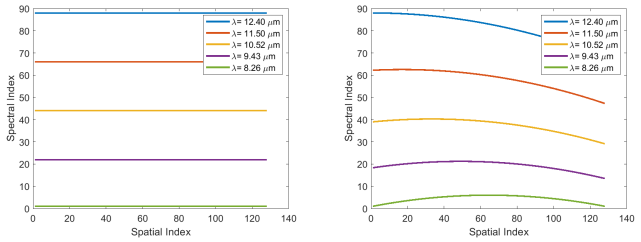


Fig. 1. Examples of (a) an ideal sensor and (b) a sensor with spatial–spectral distortion.

spectral smoothness (ARTEMISS) method [14] seek optimal atmospheric parameters from a lookup table (LUT) generated from MODTRAN [15] by minimizing residual radiance reconstruction errors. Finally, dimensionality reduction methods have been employed to learn low-dimensional linear [16] and nonlinear [17] representations of atmospheric effects from training data.

A limitation of existing TES and AC methods is that they rely on an idealized sensor model that assumes precise knowledge sensor band centers and associated bandwidths, while further assuming that these quantities are independent of spatial position within the focal plane array (FPA). In an ideal sensor, as illustrated in Fig. 1(a), a monochromatic source excites a single spectrally indexed row of the FPA that is independent of the spatial index. Whereas, a nonideal sensor exhibiting spectral-spatial distortion has a spectral response that depends on the spatial position, as seen in Fig. 1(b). Such distortions may be labeled as smile or frown [as in Fig. 1(b)]. Similarly, the optical point spread function of the system will vary as a function of field angle, which in turn will produce a variance in the bandwidth and shape of the spectral response function (SRF).

During AC and TES, if there is a discrepancy between a measurement's true wavelength λ and the sensor's assumed nominal wavelength $\tilde{\lambda}$, significant emissivity reconstruction errors may result while incorrectly attempting to invert atmospheric effects from $\tilde{\lambda}$ at the measured value of λ. Even small band center discrepancies can engender large emissivity recovery errors. For example, the results in Fig. 2 illustrate the large negative impact of band center discrepancies of 0.2 and 0.4 channel widths. Similar errors can result from uncompensated variability in spectral bandwidth.

Typically, system calibration and characterization are performed in a lab (using, e.g., a monochromator or line spectra from gas emission lamps [18]) to determine spectral band centers and associated bandwidths. However, spatial dependence is often ignored in favor of simplicity, and instability of the sensor may cause the calibrated response to drift over time due to mechanical vibrations. Repeated cooling cycles can also cause differences to exist between laboratory characterization and operating conditions in LWIR systems. Further, compact unmanned platforms are demanding lightweight and low-cost sensors, such as an uncooled microbolometer array [19]. In these settings, it is desirable for improved algorithms to compensate for low-cost or poorly calibrated sensing hardware.

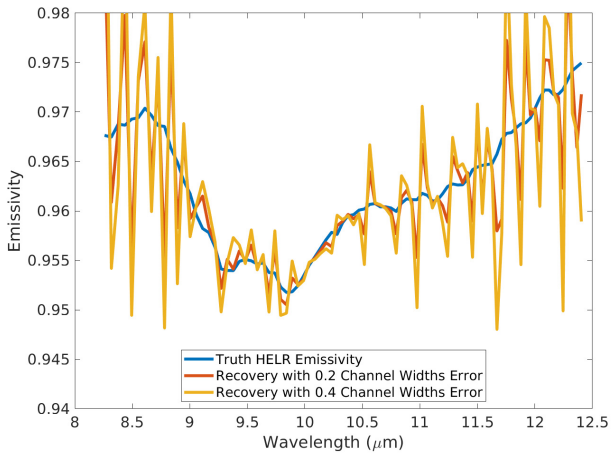


Fig. 2. Attempted emissivity recovery, in the presence of small band center distortion levels, results in large emissivity estimation errors.

In this article, we present an LWIR emissivity retrieval algorithm that simultaneously performs AC and TES while providing robustness to spectral band center and bandwidth uncertainty. Related work considers AC with spectral calibration errors in the visible and near-infrared (VNIR) and shortwave infrared (SWIR) spectral regions [20], [21]. Recent work in the LWIR domain [22] also considers a wavelength recalibration technique; however, the approach does not consider spatially dependent band center effects—such as spectral smile—and requires a spectral library to match certain in-scene materials. The primary contributions of this article are threefold:

1) A nonlinear generalization of the 2-D singular value decomposition (SVD) is developed to efficiently evaluate radiative transfer (RT) effects with the simultaneous application of unique SRFs applied to each spectral–spatial index of a sensor's 2-D FPA.
2) An estimation framework is proposed to simultaneously perform AC, TES, and spectral band center and bandwidth refinement without requiring specific in-scene components, such as blackbodies or material library elements. The algorithm leverages the above RT-SRF model in lieu of a typical LUT and utilizes a genetic algorithm for efficient optimization.
3) Experiments are performed with synthetic and measured data to quantify performance over a range of experimental conditions and against existing algorithms.

The remainder of this article is organized as follows. Section II provides a review of spectral characterization and RT models in the LWIR domain. In Section III, we present the development of a decoder network capable of efficiently modeling the 2-D matrix of atmospheric corrections applicable across the FPA. The spectrally robust AC-TES algorithm is presented in Section IV, along with the associated genetic optimization routine. Numerical results from synthetic and measured data are presented in Section V, while conclusions and future directions are given in Section VI.

II. RT and Spectral Characterization Models

In the LWIR domain, the sensor-reaching radiance at wavelength λ, corresponding to a material with emissivity $\epsilon(\lambda)$ and

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temperature T, may be modeled as [11]

$$L(\lambda) = B(\lambda, T)\epsilon(\lambda)\tau(\lambda) + (1 - \epsilon(\lambda))L_{\downarrow}(\lambda)\tau(\lambda) + L_{\uparrow}(\lambda) \tag{1}$$

where $\tau(\lambda)$ is the ground-to-sensor atmospheric transmission, $L_{\uparrow}(\lambda)$ is the atmospheric upwelling radiance, and $L_{\downarrow}(\lambda)$ is the atmospheric downwelling radiance. The function

$$B(\lambda, T) = \frac{2hc^2\lambda^{-5}}{e^{\frac{hc}{\lambda kT}} - 1} \tag{2}$$

is the spectral radiance of an ideal blackbody radiator at temperature T. In (2), h is Planck's constant, k is Boltzmann's constant, and c is the speed of light. For a hyperspectral sensor transducing N_b bands, $\{\lambda_1, \ldots, \lambda_{N_b}\}$, the measurements may be collected into a radiance vector $\boldsymbol{L} = [L(\lambda_1), \ldots, L(\lambda_{N_b})]^T$. Similarly, the per-wavelength values of the transmission, upwelling radiance, and downwelling radiance are vectorized as $\boldsymbol{\tau}$, $\boldsymbol{L}_{\uparrow}$, and $\boldsymbol{L}_{\downarrow}$, respectively. Collectively, the transmission, upwelling, and downwelling vectors make up a transmission-upwelling-downwelling (TUD) vector

$$\boldsymbol{a} = \begin{bmatrix} \boldsymbol{\tau} \\ \boldsymbol{L}_{\uparrow} \\ \boldsymbol{L}_{\downarrow} \end{bmatrix} \in \mathbb{R}^{N_t \times 1} \tag{3}$$

of length $N_t = 3\ N_b$, that completely specifies the RT effects over the bands of interest.

As described earlier, TES is generally performed after estimating the atmospheric parameters $\boldsymbol{a}$. Furthermore, if the temperature associated with a measured pixel is known, (1) may be inverted to obtain the material emissivity directly

$$\epsilon(\lambda) = \frac{L(\lambda) - L_{\downarrow}(\lambda)\tau(\lambda) - L_{\uparrow}(\lambda)}{\big(B(\lambda, T) - L_{\downarrow}(\lambda)\big)\tau(\lambda)}. \tag{4}$$

The quantity in (1) represents incident pupil-plane radiance, as dictated by materials in the scene and RT effects. However, optical distortions in the sensor give rise to nonuniform spectral and spatial scaling across the FPA. This results in uncertainty of the spectral band centers and bandwidths for each measurement acquired by the FPA. Letting k and i denote the respective spectral and spatial indicies of the FPA, the true band center at a given position may be represented as

$$\lambda_{k,i} = \tilde{\lambda}_k + \Delta_{\lambda}(k, i) \tag{5}$$

where $\tilde{\lambda}_k$ represents the nominal wavelength of band k, and $\Delta_{\lambda}(k, i)$ represents spectral–spatial distortion. The example in Fig. 2 was created by simulating a radiance measurement via the forward model (1), shifting the generative TUD vectors by w channel widths (i.e., in (5), $\Delta_{\lambda}(k, i) = w$ channel widths), and then inverting the radiance using (4) and the perturbed TUD vectors. The example highlights severe emissivity estimation errors due to band center distortion, even under the artificial scenario of a noiseless environment with known temperatures.

In practical scenarios, the distortion is often a well-behaved function of spectral and spatial position and can be modeled using the biquadratic form [11]

$$\Delta_{\lambda}(k, i) = \phi_1 x_i^2 + \phi_2 y_k^2 + \phi_3 x_i\, y_k + \phi_4 x_i + \phi_5 y_k + \phi_6 \tag{6}$$

where

$$x_i = \frac{i}{N_s}, \quad i = 1, \ldots, N_s \tag{7}$$

$$y_k = \frac{k}{N_b}, \quad k = 1, \ldots, N_b \tag{8}$$

represent scaled indicies of the spatial and spectral apertures, respectively, and the polynomial coefficients $\phi_1, \ldots, \phi_6$ control the distortion level.

In addition to uncertainty in the band centers, we also consider uncertainty in the sensor bandwidth. The latter is modeled as

$$\sigma_k = \phi_7\, \tilde{\sigma}_k \tag{9}$$

where σ_k is the actual bandwidth of band k, $\tilde{\sigma}_k$ is the nominal bandwidth of the sensor for band k, and ϕ_7 is an unknown scale factor approximately equal to one. The sensor spectral characterization parameter vector $\boldsymbol{\theta}_{\text{sc}} = [\phi_1, \ldots, \phi_7]^T$ fully describes the FPA band center locations and bandwidths necessary for data exploitation.

In view of (5), a single N_t-dimensional TUD vector cannot describe the RT effects experienced across an entire FPA because the band centers vary with the spatial position i within the array. As such, for each spatial position $i = 1, \ldots, N_s$, a TUD vector $\boldsymbol{a}_i$ can be formed based on the band centers $\lambda_{1,i}, \ldots, \lambda_{N_b,i}$ and associated bandwidths. The set of TUD vectors forms a *TUD matrix*

$$\boldsymbol{A} = [\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_{N_s}] \in \mathbb{R}^{N_t \times N_s} \tag{10}$$

that captures the RT effects and SRF applicable to all spatial–spectral positions on the FPA.

III. TUD Matrix Modeling

Letting $\boldsymbol{\theta}_{\text{atm}}$ denote a parameter vector describing the atmospheric conditions, the collection $\boldsymbol{\theta} = [\boldsymbol{\theta}_{\text{sc}}^T, \boldsymbol{\theta}_{\text{atm}}^T]^T$ jointly describes the sensor spectral characterization parameters and atmospheric parameters necessary to evaluate the TUD matrix in (10). We denote this dependence explicitly as $\boldsymbol{A}(\boldsymbol{\theta})$. In Section IV, on emissivity retrieval, it will be necessary to evaluate $\boldsymbol{A}(\boldsymbol{\theta})$ in a computationally efficient manor for a large number of candidate $\boldsymbol{\theta}$-values. RT codes, such as MODTRAN, generate high fidelity TUD vectors in a few seconds on contemporary hardware, which is acceptable for atmospheric compensation when sensing geometry and atmospheric parameters are known. However, in our case, the unknown TUD matrix $\boldsymbol{A}(\boldsymbol{\theta})$ is embedded into a loss function which must be minimized over $\boldsymbol{\theta}$, necessitating the generation of thousands of TUD vectors. Furthermore, to facilitate minimization of the loss function through numerical optimization, efficient evaluation of the derivatives of $\boldsymbol{A}(\boldsymbol{\theta})$ is essential.

To meet these objectives, we will consider low-complexity differentiable models of the TUD matrix $\boldsymbol{A}(\boldsymbol{\theta})$ that are learned from training data $\{(\boldsymbol{\theta}_i, \boldsymbol{A}_i)\}_{i=1}^N$, where $\boldsymbol{A}_i$ is the ith training TUD matrix, generated from parameters $\boldsymbol{\theta}_i$. Because training data generation and model training are offline processes, they are not required to execute quickly. As detailed in the example below, we use MODTRAN to generate training data. However, we first present three successively more complex models for efficient TUD matrix generation.

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A. Singular Value Decomposition

The SVD is frequently used to find an optimal low-dimensional approximation of a set of 1-D vectors residing in a high-dimensional subspace. In a hyperspectral imaging context, the SVD has been used to model atmospheric variability in the visible through shortwave infrared region [23], [24], and also adapted for the LWIR domain in [25], [26], and [16].

Because the SVD models 1-D data, applying these ideas to the TUD matricies $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_N$ of the present article would first require vectorizing each matrix, a process sometimes referred to as the "eigenface" approach [27]. Let $\boldsymbol{b}_i$ denote the vectorization of $\boldsymbol{A}_i$, such that each column of $\boldsymbol{A}_i$ is concatenated end-to-end to form an $(N_t N_s) \times 1$ vector. Let $\boldsymbol{B} = [\boldsymbol{b}_1, \ldots, \boldsymbol{b}_N] \in \mathbb{R}^{N_t N_s \times N}$ represent the full data matrix, and let

$$\boldsymbol{B} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{V}^T \tag{11}$$

denote its SVD. The least-squares—optimal d-dimensional subspace capturing the range of $\boldsymbol{B}$ is spanned by the first d principle left singular vectors: $\boldsymbol{U}_{1:d} = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_d]$.

The d-dimensional coefficient vector of a (vectorized) TUD matrix $\boldsymbol{b}$ is computed as

$$\boldsymbol{x} = \boldsymbol{U}_{1:d}^T \boldsymbol{b} \tag{12}$$

with the reconstruction of the vectorized TUD matrix given by

$$\boldsymbol{b} \approx \hat{\boldsymbol{b}} = \boldsymbol{U}_{1:d}\boldsymbol{x}. \tag{13}$$

The SVD is an unsupervised method, making the coefficient vector $\boldsymbol{x}$ distinct from the generative atmospheric and spectral characterization parameter vector $\boldsymbol{\theta}$. Below, we evaluate TUD matrix reconstruction performance as a function of the subspace dimension d.

B. 2-D Singular Value Decomposition

The drawbacks of using a 1-D SVD on vectorized data are twofold. First, the spatial relationships present in the TUD matrix are lost upon vectorization. In our case, we expect a strong correlation along the rows of each TUD matrix where only slight variations are expected due to small shifts in band centers. The second drawback is that, due to the large size of the data matrix $\boldsymbol{B}$ above, the SVD calculation is very memory and compute-intensive. However, this is not a serious impediment since the SVD is only performed once during offline training.

To address these issues, recent work [28] developed a 2-D extension of the SVD, making it applicable to low-dimensional representations of matrix data. Here, we review the 2-D decomposition from [28] and apply it to the collection of training TUD matrices $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_N$.

Let

$$\boldsymbol{C}_c = \sum_{i=1}^{N} (\boldsymbol{A}_i - \bar{\boldsymbol{A}})(\boldsymbol{A}_i - \bar{\boldsymbol{A}})^T \in \mathbb{R}^{N_t \times N_t} \tag{14}$$

denote the unnormalized column-wise covariance matrix, and let

$$\boldsymbol{C}_r = \sum_{i=1}^{N} (\boldsymbol{A}_i - \bar{\boldsymbol{A}})^T (\boldsymbol{A}_i - \bar{\boldsymbol{A}}) \in \mathbb{R}^{N_s \times N_s} \tag{15}$$

denote the unnormalized row-wise covariance matrix, where $\bar{\boldsymbol{A}} = (1/N)\sum_{i=1}^{N} \boldsymbol{A}_i$ is the mean of the training matricies. Forming the eigendecompositions of $\boldsymbol{C}_c$ and $\boldsymbol{C}_r$, we have

$$\boldsymbol{C}_c = \sum_{i=1}^{N_t} \xi_i \boldsymbol{u}_i \boldsymbol{u}_i^T \tag{16}$$

$$\boldsymbol{C}_r = \sum_{i=1}^{N_s} \gamma_i \boldsymbol{v}_i \boldsymbol{v}_i^T \tag{17}$$

where the orthonormal columns $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{N_t}\}$ and $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{N_s}\}$ represent the column span and row span, respectively, of the $\{\boldsymbol{A}_i\}$.

Similar to the 1-D case, we let $\boldsymbol{U}_{1:d}$ contain the first d principle eigenvectors of $\boldsymbol{C}_c$, and we let $\boldsymbol{V}_{1:s}$ contain the first s principle eigenvectors of $\boldsymbol{C}_r$. For a given TUD matrix $\boldsymbol{A}$, the 2-D decomposition coefficients are

$$\boldsymbol{X} = \boldsymbol{U}_{1:d}^T (\boldsymbol{A} - \bar{\boldsymbol{A}}) \boldsymbol{V}_{1:s} \in \mathbb{R}^{d \times s}. \tag{18}$$

The integers d and s, respectively, control the number of column-wise and row-wise basis elements used in the decomposition. The reconstructed TUD matrix is given by

$$\boldsymbol{A} \approx \hat{\boldsymbol{A}} = \boldsymbol{U}_{1:d} \boldsymbol{X} \boldsymbol{V}_{1:s}^T + \bar{\boldsymbol{A}} \tag{19}$$

and the derivative of the (i, j) reconstructed value, with respect to the input coefficients $\boldsymbol{X}$, is readily computed as

$$\frac{\partial \hat{A}_{i,j}}{\partial \boldsymbol{X}} = \tilde{\boldsymbol{u}}_i \tilde{\boldsymbol{v}}_j^T \in \mathbb{R}^{d \times s} \tag{20}$$

where $\tilde{\boldsymbol{u}}_i$ and $\tilde{\boldsymbol{v}}_i$ denote the ith row of $\boldsymbol{U}_{1:d}$ and jth row of $\boldsymbol{V}_{1:s}$, respectively. Below, we evaluate TUD matrix reconstruction performance as a function of the subspace dimensions d and s.

C. Nonlinear Decoder

Because the mapping $\boldsymbol{\theta} \mapsto \boldsymbol{A}$ is nonlinear, we further consider models based on deep neural networks (DNNs), which excel at extracting nonlinear relationships among large numbers of variables. For vector-valued data, as depicted in Fig. 3, a typical fully connected layer in a DNN maps input vector $\boldsymbol{x}^{(\ell-1)}$ to output vector $\boldsymbol{x}^{(\ell)}$ as

$$\boldsymbol{x}^{(\ell)} = f_\ell(\boldsymbol{x}^{(\ell-1)}) = \eta(\boldsymbol{W}^{(\ell)}\boldsymbol{x}^{(\ell-1)} + \boldsymbol{b}^{(\ell)}). \tag{21}$$

In (21), $\boldsymbol{W}^{(\ell)}$ and $\boldsymbol{b}^{(\ell)}$ denote a weight matrix and bias vector, respectively, that are optimally learned for layer ℓ using training data. The activation function $\eta()$ is a nonlinear function applied elementwise and imparts a nonlinear relationship between the layer's inputs and outputs. The composition of multiple layers $f_L(\ldots f_2(f_1(\boldsymbol{x}^{(0)})))$ generates a highly expressive model capable of learning complicated input-output relationships.

For nonlinear modeling of 2-D TUD matricies, we propose to use the parameterization of the 2-D SVD (19) within a feedforward neural network framework. Analogous to the vector case (21), the ℓth layer of our DNN maps input matrix $\boldsymbol{X}^{(\ell-1)}$ to output matrix $\boldsymbol{X}^{(\ell)}$ as

$$\begin{aligned} \boldsymbol{X}^{(\ell)} &= f_\ell(\boldsymbol{X}^{(\ell-1)}) \\ &= \eta(\boldsymbol{U}^{(\ell)}\boldsymbol{X}^{(\ell-1)}\boldsymbol{V}^{(\ell)^T} + \boldsymbol{B}^{(\ell)}) \in \mathbb{R}^{N_{\ell,r} \times N_{\ell,c}} \end{aligned} \tag{22}$$

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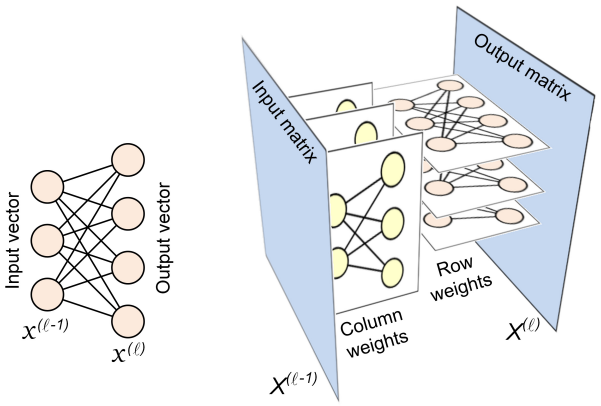


Fig. 3. Traditional layer in a fully connected feed-forward neural network (left), and the proposed per-layer structure for 2-D TUD matrix data (right).

where $\boldsymbol{U}^{(\ell)}$ and $\boldsymbol{V}^{(\ell)}$ capture the column space and row space, respectively, of the linear portion of the mapping, and $\boldsymbol{B}^{(\ell)}$ is a bias matrix for layer ℓ. The output size $(N_{\ell,r} \times N_{\ell,c})$ of layer ℓ is determined by the outer dimensions of $\boldsymbol{U}^{(\ell)}\boldsymbol{X}^{(\ell-1)}\boldsymbol{V}^{(\ell)^T}$. This implies that $\boldsymbol{U}^{(\ell)}$ is $(N_{\ell,r} \times N_{\ell-1,r})$, $\boldsymbol{V}^{(\ell)}$ is $(N_{\ell,c} \times N_{\ell-1,c})$, and $\boldsymbol{B}^{(\ell)}$ is $(N_{\ell,r} \times N_{\ell,c})$.

For an L-layer network, we have

$$\boldsymbol{X}^{(L)} = f(\boldsymbol{X}^{(0)}) = f_L(\ldots f_2(f_1(\boldsymbol{X}^{(0)}))) \tag{23}$$

where all layers are as in (22), except that the final output layer omits the activation function $\eta()$. For our TUD matrix application, and as depicted in Fig. 4, the input to the network is the physical parameter vector $\boldsymbol{X}^{(0)} = \boldsymbol{\theta}$, and the output is the TUD matrix approximation $\boldsymbol{X}^{(L)} \approx \boldsymbol{A}$. During training, we learn the optimal parameters $\boldsymbol{\Omega} = \{\boldsymbol{U}^{(\ell)}, \boldsymbol{V}^{(\ell)}, \boldsymbol{B}^{(\ell)}\}_{\ell=1}^{L}$ of the DNN by using the training data and minimizing the model's prediction error

$$\hat{\boldsymbol{\Omega}} = \arg\min_{\boldsymbol{\Omega}} \frac{1}{N}\sum_{i=1}^{N} \|\boldsymbol{A}_i - f(\boldsymbol{\theta}_i)\|_{\mathrm{F}}^2 \tag{24}$$

where $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius matrix norm. After the model parameters $\boldsymbol{\Omega}$ have been learned, forward modeling follows (23). The gradient of the (i, j)th output element of $\boldsymbol{X}^{(\ell)}$, from layer ℓ, with respect to its input $\boldsymbol{X}^{(\ell-1)}$ is

$$\frac{\partial X_{i,j}^{(\ell)}}{\partial \boldsymbol{X}^{(\ell-1)}} = \begin{cases} \eta'\left(\tilde{\boldsymbol{u}}_i^{(\ell-1)^T}\boldsymbol{X}^{(\ell-1)}\tilde{\boldsymbol{v}}_j^{(\ell-1)}\right)\tilde{\boldsymbol{u}}_i^{(\ell-1)}\tilde{\boldsymbol{v}}_j^{(\ell-1)^T} \\ \qquad\qquad 1 \le \ell \le L-1 \\ \tilde{\boldsymbol{u}}_i^{(\ell-1)}\tilde{\boldsymbol{v}}_j^{(\ell-1)^T}, \qquad \ell = L \end{cases} \tag{25}$$

where $\tilde{\boldsymbol{u}}_i^{(\ell-1)}$ and $\tilde{\boldsymbol{v}}_j^{(\ell-1)}$ denote the ith row of $\boldsymbol{U}^{(\ell-1)}$ and jth row of $\boldsymbol{V}^{(\ell-1)}$, respectively. In this work, we use the rectified linear unit (ReLU) [29] as the activation function

$$\eta(x) = \begin{cases} x, & x \ge 0 \\ 0, & x < 0 \end{cases} \tag{26}$$

and in (25), η' denotes the derivative of η.

Unlike the two unsupervised SVD approaches, the DNN model is trained in a supervised fashion with the atmospheric and spectral characterization parameters as input. Therefore, it has the advantage of having physically meaningful input

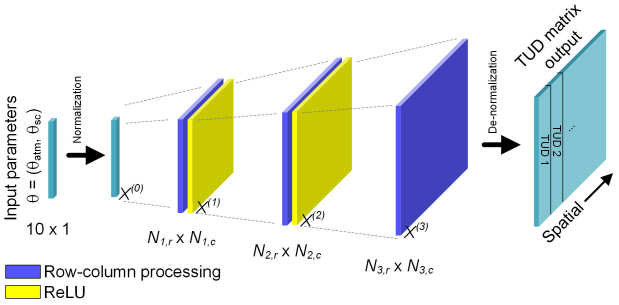


Fig. 4. A DNN decoder maps physical atmospheric parameters and sensor spectral characterization parameters to a TUD matrix that captures apparent TUD variations across the spatial dimension of the sensor's FPA.

TABLE I
RANGE OF MODTRAN PARAMETERS USED TO GENERATE TRAINING DATA. SEE TEXT FOR DETAILS

Description	Symbol	Range
Surface temperature (K)	ϕ_T	278 – 308
Surface absolute humidity (PPM)	ϕ_{H2O}	$10^2 - 10^5$
Ozone column scaling (unitless)	ϕ_{O3}	0.65 – 1.55

parameters. As detailed in the following section, the dimensionality of the atmospheric vector is $|\boldsymbol{\theta}_{\mathrm{atm}}| = 3$, making the entire input vector 10 dimensional. Because the network maps this low-dimensional input space into a high-dimensional output (below, $\boldsymbol{A} \in \mathbb{R}^{264\times128}$), we refer to the network as a decoder. This is analogous to the decoder portion of an autoencoder, which maps a low-dimensional latent space into its high-dimensional output [30].

D. Training Data and Normalization

As described earlier, we use the MODTRAN RT code to generate the training data $\{(\boldsymbol{\theta}_i, \boldsymbol{A}_i)\}_{i=1}^{N}$ necessary to train and evaluate the above three TUD matrix models. In Section V, we consider temperature emissivity separation on LWIR data cubes collected over several days from the spatially enhanced broadband array spectrograph (SEBASS) sensor [31]. Accordingly, here we generate training data consistent with the SEBASS sensor and a range of atmospheric conditions generously expected to capture the variability experienced during the data collection. Starting with the mid-latitude summer profile, as summarized in Table I, we sweep the following unknown atmospheric parameters over plausible ranges. The ground surface temperature, ϕ_T, was varied over a range of 278 K–308 K (31 uniform samples) with a constant adiabatic lapse rate of 6.5 K/km. Water vapor profiles were generated as in [10] using a range of surface absolute humidity, ϕ_{H2O}, that spanned from 10^2–10^5 parts per million volume (80 log-spaced samples). Finally, the ozone column, ϕ_{O3}, is scaled between 0.65 and 1.55 (10 uniform samples) times the default MODTRAN ozone column. Hence, the triple $\boldsymbol{\theta}_{\mathrm{atm}} = [\phi_T, \phi_{\mathrm{H2O}}, \phi_{\mathrm{O3}}]^T$ specifies a single atmosphere, while the sampling above creates a training set with a total of $N = 31\times80\times10 = 24\,800$ atmospheric conditions, $\{\boldsymbol{\theta}_{\mathrm{atm},i}\}_{i=1}^{N}$.

After removing strong atmospheric absorption bands and low SNR bands, the LWIR data available from the SEBASS sensor contain $N_b = 88$ spectral bands in the range 8.2 μm

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to 12.4 μm and $N_s = 128$ samples in the spatial dimension of the FPA. The sensor specifications provide the nominal band centers $\{\tilde{\lambda}_k\}_{k=1}^{N_b}$ and bandwidths $\{\tilde{\sigma}_k\}_{k=1}^{N_b}$. For simplicity, we draw the sensor's spatial-spectral distortion coefficients in (6) as independent uniform random variables ranging from $-c$ to c

$$\phi_i \sim \mathcal{U}(-c, c), \quad 1 \le i \le 6. \tag{27}$$

From (6), the maximum absolute distortion is then seen to be $\sum_{i=1}^{6} \max |\phi_i| = 6\ c$. It is desirable to limit the maximum distortion to a fraction of the average spectral bandwidth $\bar{\sigma}_{\mathrm{BW}}$. In our case, we choose a maximum distortion of 60% $\bar{\sigma}_{\mathrm{BW}}$, making $c = 0.1$. For the bandwidth scale factor in (9), we use a uniform random variable between 0.9 and 1.1

$$\phi_7 \sim \mathcal{U}(0.9, 1.1). \tag{28}$$

Each atmospheric condition $\boldsymbol{\theta}_{\mathrm{atm},i}$ is combined with a random realization of the sensor spectral characterization parameters $\boldsymbol{\theta}_{\mathrm{sc},i}$ to produce training element $\boldsymbol{\theta}_i = [\boldsymbol{\theta}_{\mathrm{atm},i}^T, \boldsymbol{\theta}_{\mathrm{sc},i}^T]^T$. As such, each $\boldsymbol{\theta}_i$ fully specifies the atmospheric conditions, band centers, and bandwidths across both dimensions of the FPA. This information is passed to MODTRAN to create the associated TUD matrix $\boldsymbol{A}_i$, for each $\boldsymbol{\theta}_i$. A Gaussian SRF is used in this process.

Additionally, the decoder loss function (24) considers minimization of squared errors. Because some components of a TUD vector are much larger than others, this can lead to imbalanced training. For example, downwelling radiance values, in units of μ-flicks, can be several thousand times larger than transmission loss, which has values in the range from 0 to 1. To compensate for this, we use z-score normalization such that each element of a TUD matrix is shifted and scaled to have zero mean and unit variance over the training set. Additionally, to facilitate convergence in the iterative training of the decoder model, we also apply z-score normalization to the input parameter vector $\boldsymbol{\theta}$ [32]. Letting $\mathcal{Z}()$ denote the z-score normalization operation, and $\mathcal{Z}^{-1}()$ denote denormalization, the data seen by the models is $\{\mathcal{Z}(\boldsymbol{\theta}_i), \mathcal{Z}(\boldsymbol{A}_i)\}_{i=1}^{N}$, and model outputs should be denormalized to restore the physical TUD matrix. The complete TUD matrix decoder process is illustrated in Fig. 4.

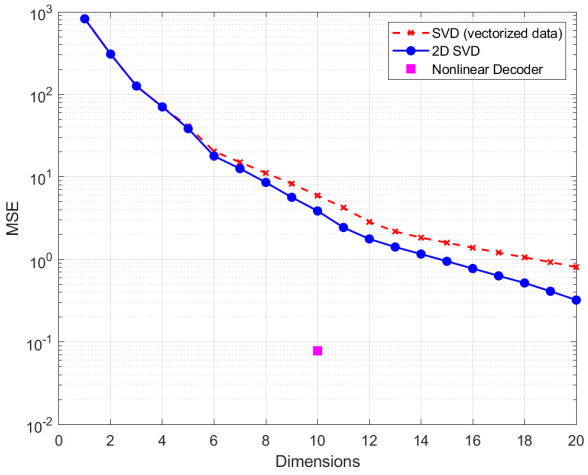


Fig. 5. TUD matrix reconstruction error versus model input dimension size. The nonlinear decoder accepts the physical parameters as inputs and thus has a fixed input size.

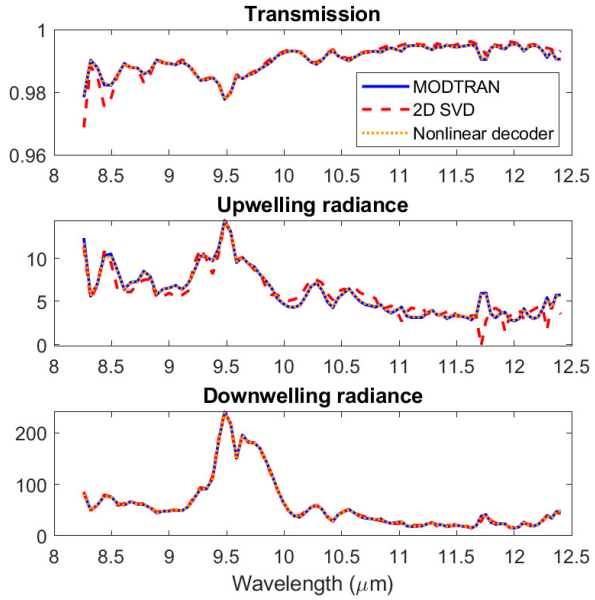


Fig. 6. Example TUD matrix reconstructions using 10-D inputs. For each method, the TUD vector from the middle of the spatial axis ($N_s/2$) of the FPA is plotted.

E. Comparison

Here, we evaluate the TUD matrix reconstruction performance of the three proposed models. In Fig. 5, we plot reconstruction mean-squared error (MSE)

$$\mathrm{MSE} = \frac{1}{N} \sum_{i=1}^{N} \|\mathcal{Z}(\boldsymbol{A}_i) - \mathcal{Z}(\hat{\boldsymbol{A}}_i)\|_{\mathrm{F}}^2 \tag{29}$$

as a function of input dimension. In (29), $\hat{\boldsymbol{A}}_i$ represents a reconstruction from any of the three methods. For the SVD, the input dimension is the size of the subspace considered—d in (12). For the 2-D SVD, an analysis of the eigenvalue spectrum of the row-space (see (17)) reveals that the majority of the row-space is captured by the first four principle eigenvectors ($\sum_{i=5}^{N_s} \gamma_i / \sum_{i=1}^{N_s} \gamma_i = 10^{-5}$). So, in (18) we set $s = 4$ and plot performance versus d in Fig. 5. For the nonlinear decoder, we consider an $L = 3$ layer implementation. Because the input is equal to the physical parameter vector $\boldsymbol{X}^{(0)} = \boldsymbol{\theta}$, the input size is fixed at $|\boldsymbol{\theta}| = 10$ in the figure. The size of the decoder's output must equal the TUD matrix dimensions, $(N_{3,r} \times N_{3,c}) = (N_t \times N_s) = (3 \cdot 88 \times 128)$, however the intermediate layer sizes are free parameters. We found $(N_{1,r} \times N_{1,c}) = (64 \times 4)$, $(N_{2,r} \times N_{2,c}) = (128 \times 16)$ to work well and used these settings throughout this article.

As seen in Fig. 5, the 2-D SVD outperforms the 1-D SVD on vectorized data, as expected. However, the reconstruction error of the nonlinear decoder is nearly two orders of magnitude lower than the 2-D SVD for 10-D inputs. Fig. 6 illustrates typical TUD vector reconstructions for the decoder and 2-D SVD method (with $d = 10$ dimensions). The vectorized 1-D

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SVD results are similar to the 2-D SVD and omitted from the figure. The methods simultaneously produce $N_s = 128$ TUD vectors, however, for illustration, only TUD vectors taken from the middle of the spatial axis ($N_s/2$) of the FPA are shown. These can be compared to the MODTRAN values. Although the 2-D SVD exhibits some artifacts, the decoder reconstruction is indistinguishable from the MODTRAN values in the figure.

While further increasing the input dimensionality of the SVD methods will necessarily lower their reconstruction error, increasing the number of parameters complicates the TUD matrix optimization process during data exploitation—such as emissivity retrieval described in the following section. The decoder also has the benefit of being parameterized by physically meaningful atmospheric and sensor spectral characterization parameters. For these reasons, we opt to use the decoder network for TUD matrix formation. Henceforth, from (23), we denote the TUD matrix decoder simply as $f(\boldsymbol{\theta}) = \boldsymbol{A}$.

IV. Emissivity Retrieval With Temperature, Atmospheric, and Band Uncertainty

Because emissivity is a unique intrinsic material property, emissivity retrieval is a common goal in LWIR signal processing applications. In this section, we utilize the TUD matrix decoder to enable emissivity retrieval in the presence of unknown ground surface temperatures, atmospheric conditions, and spectral band uncertainty in the FPA.

Existing algorithms, beginning with ISAC [10], attempt to compensate for unknown temperatures and atmospheres—but not band uncertainty. The ISAC approach estimates a single global TUD vector by performing regression on in-scene blackbodies. This yields a general TUD estimate, however, the method is limited by its reliance on finding in-scene blackbodies and can render poor TUD estimates in their absence. The ARTEMISS [14] algorithm refines and improves the ISAC estimate in these conditions.

The premise of ARTEMISS is that the correct atmosphere and material temperature estimates will produce spectrally smooth emissivity estimates. Beginning with the ISAC TUD estimate, ARTEMISS selects Q similar TUD vectors from a precomputed LUT of TUD vectors and then computes a fitness metric for each of the Q candidate TUDs. For a given TUD estimate $\{\hat{\boldsymbol{\tau}}, \hat{\boldsymbol{L}}_{\uparrow}, \hat{\boldsymbol{L}}_{\downarrow}\}$, the optimal smoothness-inducing temperature is found, for each pixel p, as

$$\hat{T}_p = \arg\min_{T_p} \ \|\boldsymbol{\ell}_p(T_p)\|_2^2 \tag{30}$$

$$\boldsymbol{\ell}_p(T_p) = \boldsymbol{L}_p - \hat{\boldsymbol{L}}_p\big(\mathcal{F}(\hat{\boldsymbol{\epsilon}}_p(\hat{\boldsymbol{\tau}}, \hat{\boldsymbol{L}}_{\downarrow}, \hat{\boldsymbol{L}}_{\uparrow}, T_p)\big) \tag{31}$$

where $\boldsymbol{L}_p$ is measured radiance, $\hat{\boldsymbol{\epsilon}}_p$ is the recovered emissivity from (4), and $\mathcal{F}$ is a convolution operator in which the emissivity is convolved with a 3-tap boxcar filter. Finally, the filtered emissivity is propagated through the forward model (1) to produce a corresponding radiance estimate $\hat{\boldsymbol{L}}_p$. The rationale behind (31) is that the optimal temperature and TUD will produce a smooth emissivity. Thus, the low-pass filter will have negligible effect, making $\hat{\boldsymbol{L}}_p$ and $\boldsymbol{L}_p$ very similar.

ARTEMISS selects the optimal TUD as the one producing the lowest error across the pixels

$$\sum_{p\in\mathcal{P}} \|\boldsymbol{\ell}_p(\hat{T}_p)\|_2^2 \tag{32}$$

where, due to high computational complexity, the evaluation pixel set $\mathcal{P}$ is chosen to be a subset of the entire scene.

A. Emissivity Retrieval Algorithm

Here, we address emissivity retrieval using the basic framework and spectral smoothness regularization found in ARTEMISS. However, the proposed approach contains the following improvements: 1) elimination of the ISAC preestimate with blackbody dependence; 2) support for fully optimized TUD vectors instead of quantized elements selected from an LUT; and 3) support for band center and bandwidth uncertainty among FPA elements.

Incorporating the spatial–spectral TUD decoder $f(\boldsymbol{\theta})$ into the spectral smoothness criteria, we obtain the residual vectors

$$\boldsymbol{\ell}_p(\boldsymbol{\theta}, T_p) = \boldsymbol{L}_p - \hat{\boldsymbol{L}}_p\big(\mathcal{F}(\hat{\boldsymbol{\epsilon}}_p(f(\boldsymbol{\theta}), T_p)\big) \tag{33}$$

$$\boldsymbol{\ell}(\boldsymbol{\theta}, \boldsymbol{T}) = \big[\boldsymbol{\ell}_1^T, \ldots, \boldsymbol{\ell}_P^T\big]^T \tag{34}$$

and associated cost function

$$c(\boldsymbol{\theta}, \boldsymbol{T}) = \sum_{p=1}^{P} \|\boldsymbol{\ell}_p\|_2^2. \tag{35}$$

The optimal set of atmospheric and spectral characterization parameters $\boldsymbol{\theta} = [\boldsymbol{\theta}_{\text{sc}}^T, \boldsymbol{\theta}_{\text{atm}}^T]^T$, and ground surface temperatures $\boldsymbol{T} = [T_1, \ldots, T_P]$, are found as the minima of this cost

$$\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{T}} = \arg\min_{\boldsymbol{\theta}, \boldsymbol{T}} \ c(\boldsymbol{\theta}, \boldsymbol{T}). \tag{36}$$

The basic optimization is performed using coordinate descent

$$\text{For } t = 1, 2, \ldots$$

$$T_p^t = \arg\min_{T_p} c\big(\boldsymbol{\theta}^{t-1}, T_1^t, \ldots, T_p, \ldots, T_P^{t-1}\big) \qquad p = 1, \ldots, P \tag{37}$$

$$\boldsymbol{\theta}^t = \arg\min_{\boldsymbol{\theta}} \ c\big(\boldsymbol{\theta}, T_1^t, \ldots, T_P^t\big) \tag{38}$$

which has the advantage that, given a $\boldsymbol{\theta}$-parameter estimate, the joint optimization over the ground surface temperatures decouples into P scalar optimizations. The individual minimizations in (37) and (38) are performed using Gauss–Newton optimization with step sizes δ_T and δ_θ, respectively. In our experiments, we use $\delta_T = \delta_\theta = 1$ throughout. The Gauss–Newton steps are efficiently implemented due to the availability of the closed form derivatives of the TUD matrix decoder and the cost function components. These derivatives are provided in the Appendix.

To combat nonconvexity in the cost, we utilize a greedy genetic optimization routine where Q initial parameters $\boldsymbol{\Omega} = \{\boldsymbol{\theta}_1^0, \ldots, \boldsymbol{\theta}_Q^0\}$ are randomly chosen between bounds $\boldsymbol{\theta}_{\min}$ and $\boldsymbol{\theta}_{\max}$. In all of our experiments, we found $Q = 128$ to be sufficient. As the optimization proceeds, each member of the $\boldsymbol{\Omega}$ population evolves toward the minimal cost, with only half of the candidate solutions surviving at each iteration. Performing $\lceil \log_2(Q) \rceil$ iterations ensures that only the single most optimal

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candidate survives, while an optional N_{it} additional iterations further refine this solution. In our experiments, we found $N_{\text{it}} = 5$ to perform well, however other pruning and stopping criteria are design alternatives. The complete optimization process is detailed in Algorithm 1, which we refer to as Decoding Infrared Atmospheres iN ARTEMISS or DIANA.

The final two inputs to the algorithm are the evaluation pixels $\mathcal{P}$ and the initial temperature values. For the initial temperatures, we use the mean of an upper temperature bound $T_{\max}$ and the maximum between a minimum temperature bound $T_{\min}$ and the maximum apparent spectral temperature (AST) [3] of each pixel $T_p^{(\text{AST})}$

$$T_p^0 = \frac{1}{2}\left(\max\left(T_{\min}, T_p^{(\text{AST})}\right) + T_{\max}\right). \tag{39}$$

In selecting a subset $\mathcal{P}$ of the total pixels for evaluation, it is desirable to select a spectrally diverse set in order to minimize measurement redundancy. While numerous selection strategies are possible, we find that vertex component analysis (VCA) [33] is both computationally efficient and effective in the DIANA algorithm.

Algorithm 1 DIANA Algorithm

Input: $\boldsymbol{L}$, T_{min}, T_{max}, $\boldsymbol{\theta}_{min}$, $\boldsymbol{\theta}_{max}$, δ_T, $\delta\theta$, Q, P, N_{it}
Output: $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{T}}$, $\hat{\boldsymbol{\epsilon}}$

Using VCA, select P evaluation pixels from $\boldsymbol{L}$
Randomly sample Q vectors in the range $[\boldsymbol{\theta}_{min}, \boldsymbol{\theta}_{max}]$ to form the initial candidate parameter set $\Omega = \{\boldsymbol{\theta}_1^0, \ldots, \boldsymbol{\theta}_Q^0\}$.
for $t = 1$ *to* $\lceil log_2(Q)\rceil + N_{it}$ **do**
  for $r = 1$ *to* $|\Omega|$ **do**
    Calculate the TUD matrix across entire FPA using the decoder (23), $\boldsymbol{A}_r = f(\boldsymbol{\theta}_r^{t-1})$.
    For each pixel $p = 1, \ldots, P$, calculate the residual $\boldsymbol{\ell}_p(\boldsymbol{A}_r, T_p^{t-1})$ via (33).
    Compute the Jacobian, $\boldsymbol{J}_{T_p}$, of $\boldsymbol{\ell}_p$ with respect to T_p via (48).
    For each pixel, p, update the temperature estimate using Gauss–Newton optimization

$$T_p^t = T_p^{t-1} - \delta_T\left(\boldsymbol{J}_{T_p}^T \boldsymbol{J}_{T_p}\right)^{-1} \boldsymbol{J}_{T_p}^T \boldsymbol{\ell}_p \tag{40}$$

    projecting values outside $[T_{min}, T_{max}]$ to the boundary.
    Compute the total residual vector $\boldsymbol{\ell}(\boldsymbol{A}_r, \boldsymbol{T}^t)$ via (34).
    Compute the Jacobian, $\boldsymbol{J}_\theta$, of $\boldsymbol{\ell}(\boldsymbol{\theta}, \boldsymbol{T})$ with respect to $\boldsymbol{\theta}$ via (51).
    Update $\boldsymbol{\theta}$ using Gauss–Newton optimization

$$\boldsymbol{\theta}_r^t = \boldsymbol{\theta}_r^{t-1} - \delta_\theta\left(\boldsymbol{J}_\theta^T \boldsymbol{J}_\theta\right)^{-1} \boldsymbol{J}_\theta^T \boldsymbol{\ell} \tag{41}$$

    projecting values outside $[\boldsymbol{\theta}_{min}, \boldsymbol{\theta}_{max}]$ to the boundary.
    Calculate total loss $c_r = c(\boldsymbol{\theta}_r^t, \boldsymbol{T}_r^t)$ via (35).
  Using the losses in Step 12, remove the worst half of the candidate $\boldsymbol{\theta}$'s from Ω, retaining at least one $\boldsymbol{\theta}$.
Return final temperatures $\{T_p\}$ and parameter vector $\boldsymbol{\theta}$, along with final emissivity estimates via (4).

B. DIANA Special Cases

If the sensor has been carefully calibrated and operating conditions are stable, band uncertainty among the FPA elements may be insignificant. In this case, the spectral characterization parameters $\boldsymbol{\theta}_{\text{sc}}$ may be removed from the parameter vector leaving only temperature and atmospheric uncertainty, $\boldsymbol{\theta} = \boldsymbol{\theta}_{\text{atm}}$. This reduces the complexity of the decoder and minimizes the number of parameters to be estimated, resulting in computational and performance improvements.

A second special case is when the band centers and bandwidths are unknown, but are constant across the spatial dimension of the FPA. This scenario accommodates spectral calibration errors, but not spatial–spectral distortion such as smile, which manifests as spatially varying spectral sensitivity. In this case, (6) becomes

$$\Delta_\lambda(k) = \phi_2 y_k^2 + \phi_5 y_k + \phi_6 \tag{42}$$

and the spectral characterization parameters reduce to $\boldsymbol{\theta}_{\text{sc}} = [\phi_2, \phi_5, \phi_6, \phi_7]^T$, with total parameter vector $\boldsymbol{\theta} = [\boldsymbol{\theta}_{\text{atm}}^T, \boldsymbol{\theta}_{\text{sc}}^T]^T$. As in the previous special case, this has the advantage of reducing the decoder complexity and total number of parameters to be estimated.

TABLE II
TIME AND CONDITIONS OF MEASURED-DATA EXPERIMENTS

Experiment	mm/dd/yyyy	Time of day	Weather
1	08/12/2014	Morning	Clear
2	08/12/2014	Morning	Clear
3	08/12/2014	Morning	Clear
4	08/12/2014	Afternoon	Clear
5	08/12/2014	Afternoon	Clear
6	08/12/2014	Afternoon	Clear
7	08/18/2014	Afternoon	Clear
8	08/13/2014	Afternoon	Cloudy
9	08/11/2014	Afternoon	Clear
10	08/19/2014	Night	Clear
11	08/24/2014	Afternoon	Cloudy

V. RESULTS

In this section, we evaluate the effectiveness of the DIANA algorithm on both measured and synthetic data. The measured data are derived from the SEBASS [31] sensor flown at an altitude of 1500 ft above ground level (AGL). After removing strong atmospheric absorption bands and low SNR bands, the data contain $N_b = 88$ bands in the 8.3–12.4 μm range and $N_s = 128$ cross-track spatial pixels. As described in Table II, multiple data cubes of a common scene were collected over a 14-day period. The eleven experiments offer variety in temperature, humidity, time of day, and cloud conditions to assess the robustness of the developed method. These conditions, along with the nominal SEBASS band centers and bandwidth specifications, give rise to the training data parameter sampling detailed in Section III-D. The resulting TUD decoder is used in both the synthetic and measured results below.

A. Synthetic Data

The objective of the first experiment is to evaluate the robustness of the DIANA algorithm to spectral band center and bandwidth uncertainty. In this test, a random atmosphere is created by randomly sampling the ozone, water vapor,

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TABLE III

NESR ERROR RATES (IN μ-FLICKS) OF THE PROPOSED ALGORITHM FOR DIFFERENT DISTORTION TYPES

Algorithm \ *Distortion*	None	Spect.	Spect. & spatial	Time (s)
No distortion correction	**0.90**	2.61	3.24	0.13
Spectral correction	1.01	**1.03**	1.32	0.15
Spect. & spatial corr.	1.16	1.15	**1.16**	0.48

and ground temperature values from the ranges described in Section III-D. Next, we simulate sensor optical distortions by applying one of the models described earlier: 1) full spatial–spectral distortion; 2) only spectral distortion; and 3) no distortion. The distortions were created such that the band centers do not vary by more than 60% of the average SEBASS channel width. Thus, simulated distortions are within the limits of the trained decoder in Section III-D. Synthetic scenes were generated by randomly populating an $M = 2560$ pixel area from an emissivity library and randomly choosing pixel-dependent ground surface temperatures between 300 K and 330 K. Finally, all of these components are combined using the at-sensor radiance model (1) to generate hyperspectral data cubes, which included additive white Gaussian noise (AWGN) with a standard deviation of 0.5 μ-flicks.

Next, the DIANA algorithm was applied to each simulated data cube to recover the TUD vectors. The recovered TUD vectors are used with the known temperatures and known spectral emissivity to calculate the effective measurement noise induced by incorrect TUD vector estimates, i.e., the noise-equivalent spectral radiance (NESR) [11]

$$\text{NESR} = \left(\frac{1}{N_b M} \sum_{p=1}^{M} \| \boldsymbol{L}_p - \hat{\boldsymbol{L}}_p(\hat{\boldsymbol{A}}) \|_2^2 \right)^{1/2} \tag{43}$$

where $\boldsymbol{L}_p$ denotes the true radiance vector of pixel p, and $\hat{\boldsymbol{L}}_p$ is the reconstructed value from (1) using the estimated TUD matrix $\hat{\boldsymbol{A}}$. These results are averaged over 100 000 trials to produce the error values presented in Table III. In addition to the general DIANA algorithm, we also test the special cases described in Section IV-B: no distortion correction and only spectral distortion correction.

From the table, we observe that when the measurements contain both spectral and spatial distortion, not accounting for these effects results in a very large increase in the effective error: an NESR value of 3.24 μ-flicks, which is 6.5$\times$ the sensor noise of 0.5 μ-flicks. When the most general form of the proposed algorithm is used (containing both spectral and spatial distortion compensation), the error is reduced to 1.16 μ-flicks, or only 2.3$\times$ the sensor noise.

In general, we see the expected result that the best performance is obtained when the algorithm's distortion correction model matches the true distortion type (bold entries in Table III). When the algorithm is over- or under-parameterized, performance is degraded. However, using the most general spatial and spectral correction approach (bottom row of Table III) incurs a minimal NESR increase for lesser distortions, while producing a large error reduction when both distortion types are present.

TABLE IV

ESTIMATION PERFORMANCE UNDER SPECTRAL AND SPATIAL DISTORTION. ROWS 1–3 REPORT NORMALIZED RMS ERROR RATES FOR THE THREE TUD RADIOMETRIC QUANTITIES, AND ROW 4 REPORTS RMS TEMPERATURE ERRORS

		Correction type		
#	Component	None	Spect.	Spect.-spatial
1	Transmission error (dB)	-42.4	-51.5	-53.5
2	Upwelling error (dB)	-20.6	-28.7	-30.0
3	Downwelling error (dB)	-22.9	-29.3	-29.9
4	Temperature error (K)	1.88	1.68	1.60

The last column in Table III presents the average scene-wide emissivity reconstruction times, as computed on an i7-7700HQ 2.8 GHz CPU. As expected, runtime increases with model complexity. However, even the most general algorithm executes in less than half a second.

Next, we evaluate the temperature and TUD reconstruction errors within the DIANA algorithm. To isolate the errors associated with each of the three TUD radiometric quantities, we consider the normalized root mean-squared error (NRMSE)

$$\text{NRMSE} = \left(\frac{1}{M} \sum_{m=1}^{M} \frac{\| \boldsymbol{v}_m - \hat{\boldsymbol{v}}_m \|^2}{\| \boldsymbol{v}_m \|^2} \right)^{\frac{1}{2}} \tag{44}$$

where vector $\boldsymbol{v}$ represents one of $\boldsymbol{\tau}$, $\boldsymbol{L}_{\uparrow}$, or $\boldsymbol{L}_{\downarrow}$, and $\hat{\boldsymbol{v}}$ is the estimate of $\boldsymbol{v}$. In computing (44), we use the simulated data described above that includes both spectral and spatial distortion, and we average the performance over all pixels and all 100 000 simulated scenes of randomized atmospheric conditions and emissivities. In Table IV, rows 1–3, we report the NRMSE for each of the three TUD componnents in decibels (dB): $20 \log_{10}(\text{NRMSE})$. In Table IV, row 4, we also report the RMSE of the material temperature estimates

$$\left(\frac{1}{M} \sum_{m=1}^{M} (T_m - \hat{T}_m)^2 \right)^{\frac{1}{2}} \tag{45}$$

in kelvin.

As with earlier results, we observe increasing estimation performance as the compensation mechanism of DIANA is improved from no spectral–spatial correction, to spectral-only correction, to spectral and spatial corrections. In going from no correction to spectral–spatial correction, the TUD errors are reduced by 6 dB (a factor of two) or more. The temperature error reduction is more modest, however, this is to be expected as the additional modeling components in the nonlinear decoder are specifically designed to capture TUD distortions.

Finally, we evaluate the sensitivity of each algorithm to the severity of spectral band center variability. The variability of the band center distortion is quantified as

$$\text{variability} = \sqrt{\frac{1}{\bar{\sigma}^2 N_s N_b} \sum_{i=1}^{N_s} \sum_{k=1}^{N_b} \Delta_\lambda(k,i)^2} \tag{46}$$

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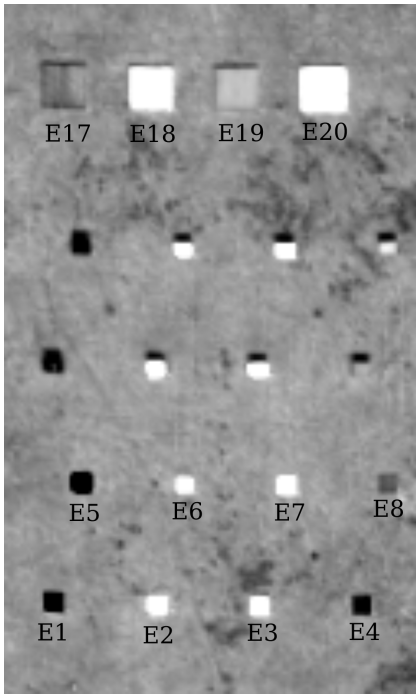


Fig. 7. Material layout of the measured data experiments.

where, from (5), $\Delta_\lambda(k,i)$ represents the spectral deviation of the (k,i) FPA element from the sensor's nominal band center $\tilde{\lambda}_k$. The normalization in (46) is chosen such that the variability is expressed as a multiple of the sensor's average channel width, $\bar{\sigma}$.

Through simulation, we generate random scenes with random realizations of the atmospheric parameters $\boldsymbol{\theta}_{\text{atm}}$ and spectral characterization parameters $\boldsymbol{\theta}_{\text{sc}}$. Different realizations of $\boldsymbol{\theta}_{\text{sc}}$ induce varying levels of spectral-spatial band center variability, and in Fig. 9 we plot the NESR performance of each variant of DIANA versus this variability.

As a control case, we consider the NESR of an algorithm that has perfect knowledge of the temperature and atmospheric parameters, but does not attempt to compensate for band center variability. When the spectral–spatial variability is low, the control naturally performs well. However, performance deteriorates when spectral–spatial variability is introduced. The DIANA variant not performing spectral–spatial calibration naturally performs worse than the control due to also having to estimate temperature and atmospheric parameters. However, despite having to estimate temperature and atmospheric conditions, DIANA with spectral-only correction achieves performance similar to the control. By accommodating both spectral and spatial distortion, the full DIANA algorithm is largely unaffected by the distortion and outperforms the control for even modest values of variability. At 0.25 channel widths of distortion, the uncompensated error is 10.3 μ-flicks while the full DIANA algorithm has 1.7 μ-flicks of error—a 6.1$\times$ reduction, and comparable to the sensor noise. This experiment demonstrates the desired robustness to sensor calibration uncertainty in the emissivity retrieval process.

B. Measured SEBASS Data

In this section, we evaluate the proposed algorithm on the 11 measured SEBASS data cubes described above and make comparisons to existing approaches. Fig. 7 displays the target array used during the measurements [34]. We exclude targets E3, E6, and E9 to E16 as they are duplicates of other materials. In addition, we exclude E5 due to a lack of emissivity truth. The lab-measured emissivity profiles of the nine remaining targets are depicted in Fig. 8. These are taken as ground truth in our experiments.

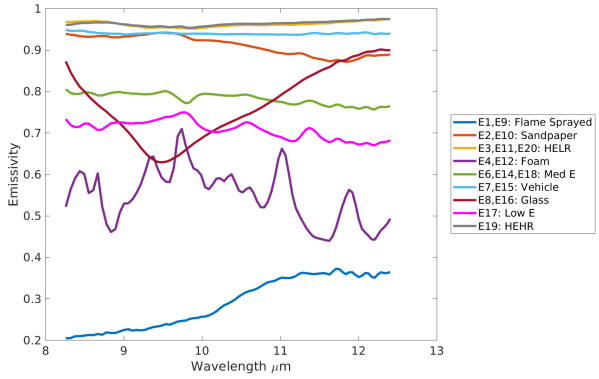


Fig. 8. Labels and spectral emissivity truth of the nine materials in the measured data.

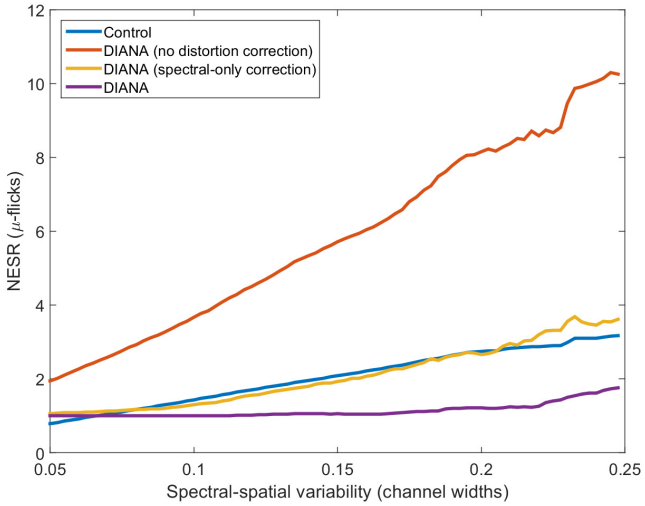


Fig. 9. Algorithm performance as a function of spectral–spatial distortion level.

The DIANA algorithm with full spatial–spectral distortion correction is applied to the datacubes, and for comparison, the ARTEMISS and FLAASH-IR algorithms are applied to the same data. Emissivity estimates for all nine material targets are shown in Fig. 10, where we observe that, qualitatively, the DIANA estimates appear to do a better job capturing the spectral shape and estimating the proper locations of peaks and valleys. Additionally, artifacts of the ozone absorption band at 9.6 μm [11] remain evident in the ARTEMISS and FLAASH-IR reconstructions, while the DIANA estimates appear properly compensated.

In Table V, we quantify the recovered emissivity errors for the different material types in the scene while averaging over the 11 experimental conditions given in Table II. In Table VI, emissivity reconstruction performance is averaged over the materials and broken out by the experimental conditions.

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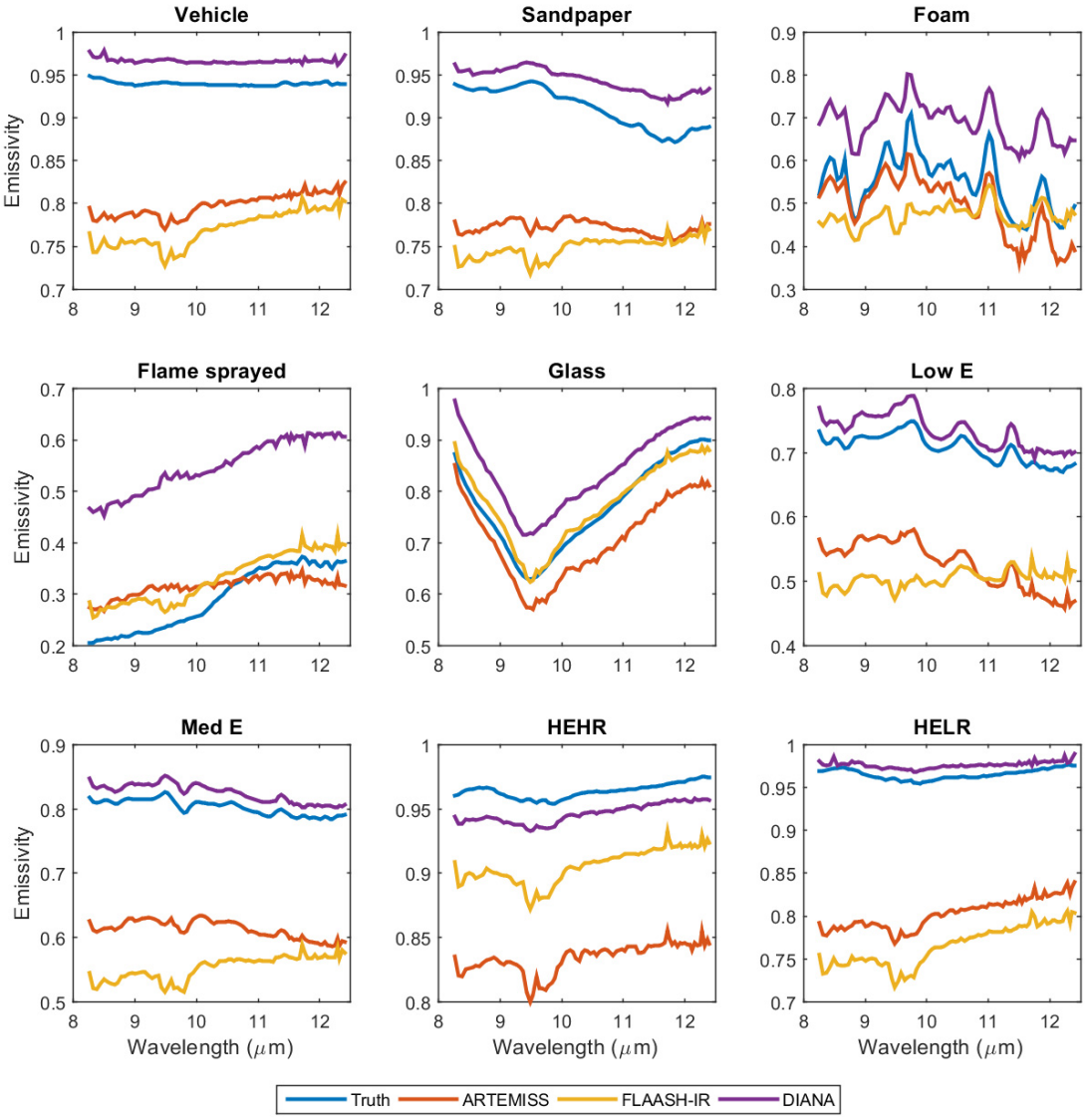


Fig. 10. Recovered target emissivities from measured SEBASS data.

The errors reported in Tables V and VI are root-mean-squared emissivity estimation errors after mean subtraction

$$\mathrm{RMSE} = \left(\frac{1}{N_b} \sum_{k=1}^{N_b} [(\epsilon_k - \bar{\epsilon}) - (\hat{\epsilon}_k - \bar{\hat{\epsilon}})]^2 \right)^{1/2} \tag{47}$$

where $\bar{\epsilon}$ represents the spectral average emissivity of a subject material, and $\bar{\hat{\epsilon}}$ represents the spectral average of the emissivity estimate $\hat{\epsilon}$ of a given material.

We consider the mean-subtracted metric in order to quantify errors in the spectral shape of the estimated emissivity vector—without regard to a constant offset across all bands. In practice, small errors in ground surface temperature estimates produce such offsets [35], and it is therefore desirable to quantify error in spectral shape—as would be used in subsequent material classification or change detection, for example.

These results show DIANA improvements over existing algorithms with better or competitive performance on almost all materials and environmental conditions considered. Considering average performance across all materials and experiments, the proposed DIANA algorithm has an error that is 26% lower than ARTEMISS, and 31% lower than FLAASH-IR.

TABLE V
EMISSIVITY ESTIMATE RMSE FROM MEASURED DATA, ORGANIZED BY MATERIAL

	Method		
Material	ARTEMISS	FLAASH-IR	DIANA
Flame	0.0315	**0.0188**	0.0212
Sandpaper	0.0239	0.0257	**0.0147**
Foam	0.0386	0.0484	**0.0328**
Vehicle	0.0152	0.0157	**0.0077**
Glass	**0.0250**	0.0259	0.0269
Low E	0.0148	0.0203	**0.0106**
Med E	0.0114	0.0164	**0.0085**
HEHR	0.0117	0.0124	**0.0068**
HELR	0.0152	0.0151	**0.0080**
Mean	0.0208	0.0221	**0.0153**

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TABLE VI
EMISSIVITY ESTIMATE RMSE FROM MEASURED DATA, ORGANIZED BY EXPERIMENT IN TABLE II

	Method		
Experiment	ARTEMISS	FLAASH-IR	DIANA
1	0.0200	0.0215	**0.0118**
2	0.0195	0.0254	**0.0113**
3	0.0193	0.0250	**0.0107**
4	0.0202	0.0245	**0.0139**
5	0.0199	0.0256	**0.0108**
6	0.0194	0.0258	**0.0121**
7	0.0187	**0.0118**	0.0119
8	0.0255	**0.0173**	0.0230
9	0.0171	**0.0116**	0.0122
10	0.0265	0.0322	**0.0254**
11	0.0222	**0.0221**	0.0229
Mean	0.0208	0.0221	**0.0153**

VI. CONCLUSION

In this article, we considered the problem of emissivity retrieval from LWIR data cubes. In addition to unknown ground surface temperatures and unknown atmospheric conditions, we paid particular attention to band center and bandwidth uncertainty that typically exists in most systems due to spatial–spectral distortion. In this setting, performing atmospheric compensation necessitated the formation of a TUD matrix, with each elemental TUD vector applying to a single spatial position on the FPA.

One challenge in estimating a large number of nuisance parameters was the need to evaluate numerous TUD matrices while minimizing the DIANA objective function. To this end, we proposed a nonlinear generalization of the 2-D SVD to model TUD matrices. The differentiability and low computational complexity of the resulting TUD matrix decoder were enabling components of the resulting DIANA algorithm. For example, the evaluation of our MATLAB code on 2.8 GHz i7-7700HQ CPU took 0.19 ms to compute a TUD matrix with $N_s = 128$ spatial components. This equates to the generation of 6.7×10^5 TUD vectors per second. Combining the decoder within a genetic optimization framework yielded an effective solution to the high-dimensional emissivity estimation problem.

As designed, the proposed DIANA algorithm demonstrated significant robustness to unknown band center variability within the FPA. Simulation results highlighted a $6.1\times$ reduction in the effective error rate of data with spatial–spectral distortion. The algorithm was also evaluated on measured data across a number of materials and a range of environmental conditions spanning many days. Compared to the ARTEMISS and FLAASH-IR algorithms, the proposed algorithm had an average error reduction of 26% and 31%, respectively.

The exploitation of LWIR measurements is frequently difficult due to a large number of unknown nuisance parameters. Future work will leverage the proposed algorithm's ability to efficiently estimate temperature, atmospheric components, and sensor spectral characterization components in order to provide spectral robustness to downstream exploitation tasks, such as material classification and change detection.

APPENDIX
DERIVATIVES

The derivatives referenced for the Gauss–Newton steps in Algorithm 1 are presented here. The Jacobian of the residual vector with respect to temperature is seen to be

$$\frac{\partial \ell_{p,k}}{\partial T_p} = -\mathcal{F}_k\left(\frac{\partial \boldsymbol{\epsilon}_p}{\partial T_p}\right)\big(B(T_p,\lambda_k) - L_{\downarrow,k}\big)\tau_k + \frac{\partial B(T,\lambda_k)}{\partial T}\mathcal{F}_k\big(\boldsymbol{\epsilon}_p\big)\tau_k \quad (48)$$

where

$$\frac{\partial B(T,\lambda_k)}{\partial T} = \frac{2hc^2}{\lambda_k^5}\frac{hc}{k_B\lambda_k}\frac{\exp\left(\frac{hc}{k_B\lambda_k T}\right)}{\left(\exp\left(\frac{hc}{k_B\lambda_k T}\right)-1\right)^2} \quad (49)$$

is the derivative of the blackbody radiance in (2), and

$$\frac{\partial \epsilon_{p,k}}{\partial T_p} = -\frac{\epsilon_{p,k}\frac{\partial B(T_p,\lambda_k)}{\partial T_p}}{B(T_p,\lambda_k) - L_{\downarrow,k}}. \quad (50)$$

In (48) and below, $\mathcal{F}_k(\boldsymbol{v})$ denotes the kth element of vector $\boldsymbol{v}$ after convolving with a 3-tap boxcar filter.

The elements of the Jacobian of the residual vectors with respect to the $\boldsymbol{\theta}$-parameters can be shown to be

$$\begin{aligned}\frac{\partial \ell_{p,k}}{\partial \theta_w} &= \tau_k\left(\mathcal{F}_k\left(\frac{\partial \boldsymbol{\epsilon}_p}{\partial \theta_w}\right)\big(B(T,\lambda_k) - L_{\downarrow_k}\big) - \big(\mathcal{F}_k\big(\boldsymbol{\epsilon}_p\big)-1\big)\frac{\partial L_{\downarrow_k}}{\partial \theta_w}\right)\\ &\quad + \mathcal{F}_k\big(\boldsymbol{\epsilon}_p\big)\frac{\partial \tau_k}{\partial \theta_w}\big(B(T,\lambda_k) - L_{\downarrow,k}\big) + L_{\downarrow,k}\frac{\partial \tau_k}{\partial \theta_w} + \frac{\partial L_{\uparrow,k}}{\partial \theta_w}\end{aligned} \quad (51)$$

where

$$\begin{aligned}\frac{\partial \epsilon_{p,k}}{\partial \theta_w} &= \frac{\frac{\partial L_{\downarrow,k}}{\partial \theta_w}(L_{p,k} - L_{\downarrow,k}(\theta)\tau_k(\theta) - L_{\uparrow,k}(\theta))}{(B(T,\lambda_k) - L_{\downarrow,k}(\theta))^2\tau_k(\theta)}\\ &\quad - \frac{\frac{\partial \tau_k}{\partial \theta_w}(L_{p,k} - L_{\downarrow,k}(\theta)\tau_k(\theta) - L_{\uparrow,k}(\theta))}{(B(T,\lambda_k) - L_{\downarrow,k}(\theta))\tau_k^2(\theta)}\\ &\quad - \frac{\tau_k(\theta)\frac{\partial L_{\downarrow,k}}{\partial \theta_w} + L_{\downarrow,k}(\theta)\frac{\partial \tau_k}{\partial \theta_w} + \frac{\partial L_{\uparrow,k}}{\partial \theta_w}}{(B(T,\lambda_k) - L_{\downarrow,k}(\theta))\tau_k(\theta)}.\end{aligned} \quad (52)$$

Finally, the derivatives of the decoder's TUD matrix outputs, with respect to the input parameters, are given by application of the chain rule to (25).

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